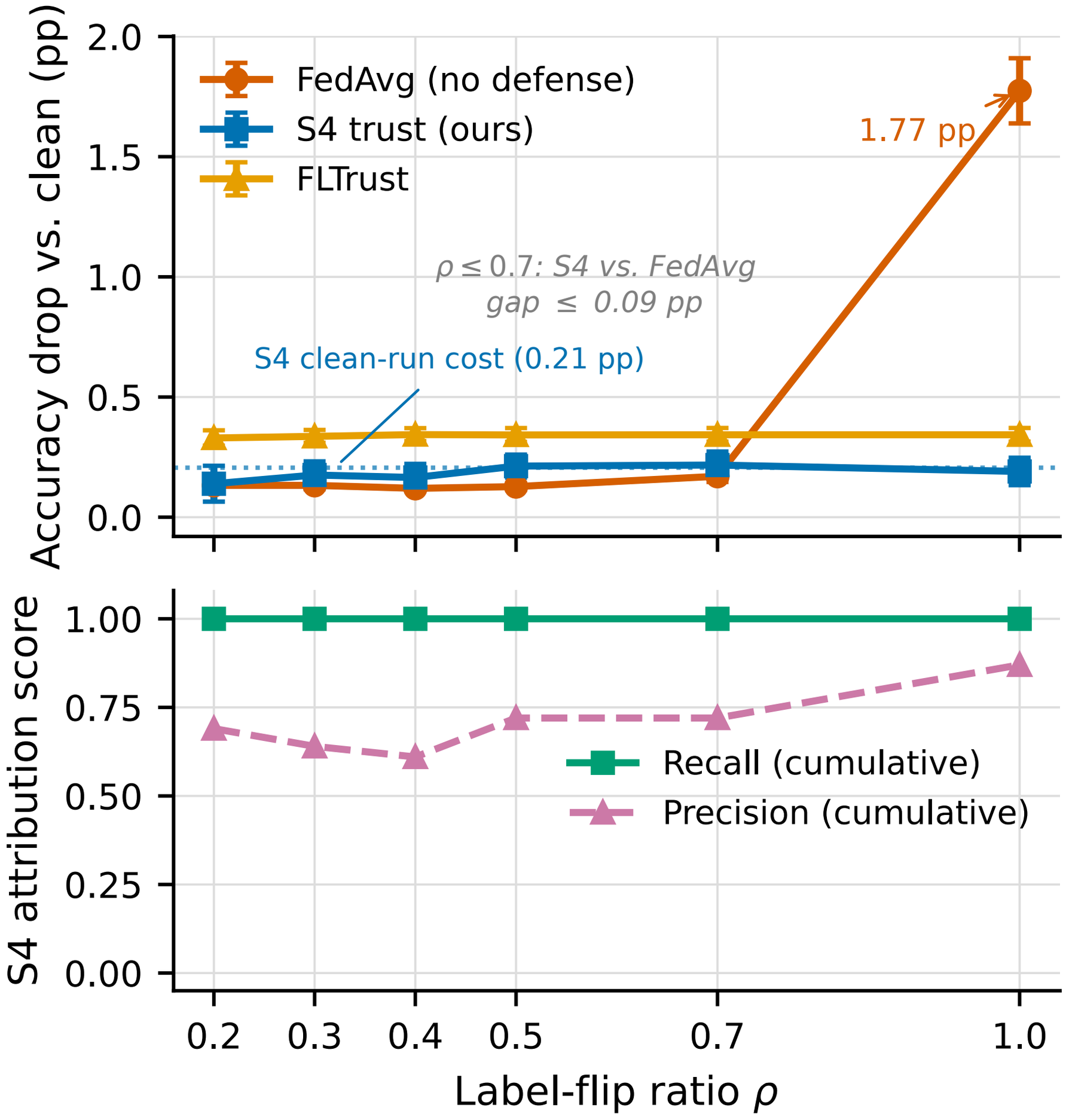
Accuracy drop vs. clean (pp)
2.0
1.5
1.0
0.5
0.0
FedAvg (no defense)
S4 trust (ours)
FLTrust
1.77 pp
ρ ≤ 0.7: S4 vs. FedAvg
gap ≤ 0.09 pp
S4 clean-run cost (0.21 pp)
S4 attribution score
1.00
0.75
0.50
0.25
0.00
Recall (cumulative)
Precision (cumulative)
0.2
0.3
0.4
0.5
0.7
1.0
Label-flip ratio ρ